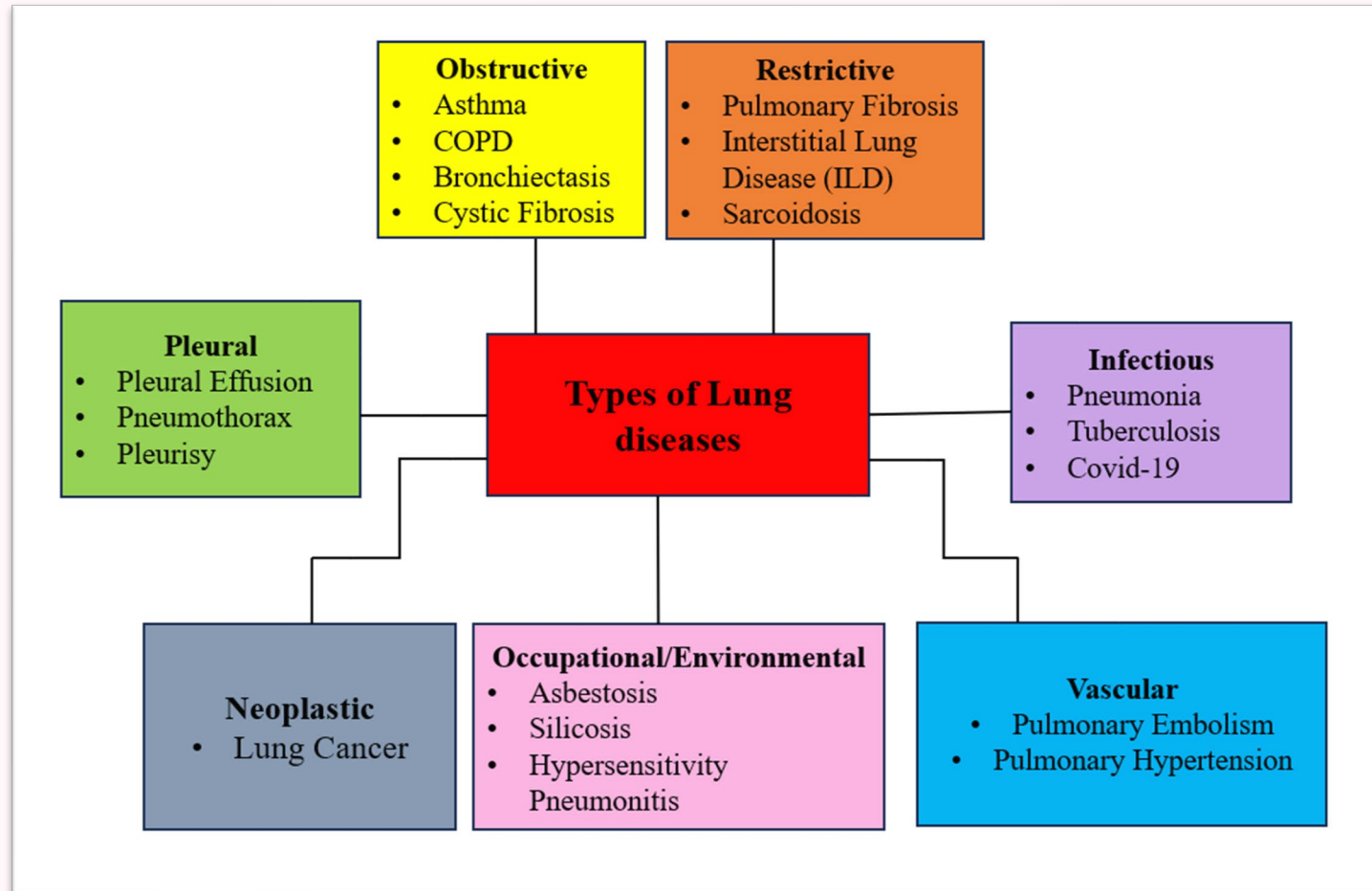
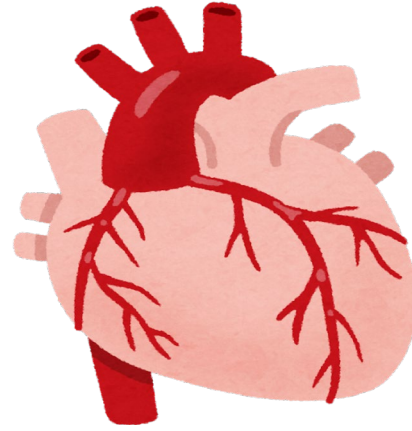
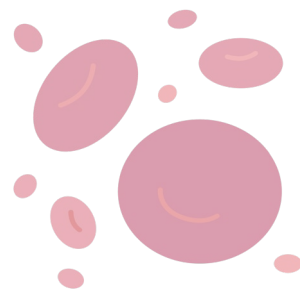
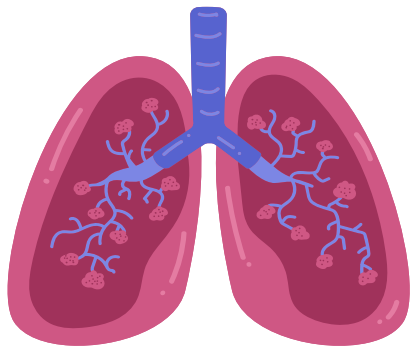


Introduction

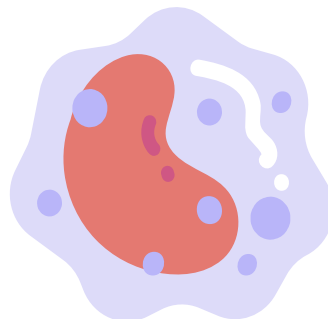
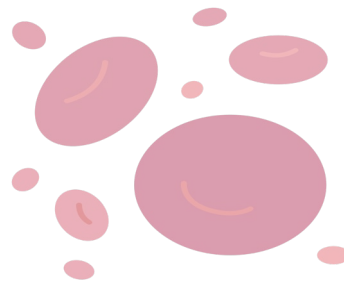
- In today's world millions of people are affected by **lung diseases** by which quality of life is reduced.
- Advanced tools
 - Spirometry,
 - laboratory pulmonary test
 - CT imaging
- highly trained specialists and high-cost equipment
- **lung beats**
- lung sounds are sliced into 2-3s clips corresponding to
 - **inspiration**
 - **expiration**
 - **short acoustic events.**
- Automatic lung sound prediction system
- The proposed work focuses on various types of lung diseases, feature extraction techniques and CNN architectures.

Lung disease Categories





Literature Survey



Papers 1-5

Sl No.	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
1	Perna et al.	2019	Respiratory anomalies	Deep learning on audio	ICBHI	CNN	Noise sensitivity	Noise-robust models
2	Rocha et al.	2019	Respiratory diseases	Dataset creation	ICBHI	—	Limited diversity	Larger real-world datasets
3	Demir et al.	2020	Lung diseases	Feature + DL	ICBHI	CNN	Complex dependency	Simplified architectures
4	Grzywalski et al.	2019	Lung sound classes	DL implementation	ICBHI	CNN	Limited clinical validation	Real-time clinical testing
5	Serbes et al.	2018	Respiratory disorders	Time-frequency	ICBHI	CNN	Computational cost	Efficient models

Papers 6-10

Sl No.	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
6	kim et al.	2021	Lung abnormalities	Transfer learning	ICBHI	CNN	Overfitting	Better augmentation
7	Bae et al.	2021	COPD, Asthma	Spectrogram-based	ICBHI	CNN	Limited features	Hybrid features
8	Pham et al.	2020	Respiratory classes	Robust DL	ICBHI	CNN	Generalization	Cross-dataset validation
9	Pramono et al.	2017	Respiratory analysis	Systematic review	Multiple	—	No implementation	Practical DL models
10	Gurung et al.	2011	Respiratory diseases	Signal analysis	Clinical	—	No DL	DL-based automation

Papers 11-15

Sl.No	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
11	Gupta et al.	2021	COPD	VMD+Gamma tone	ICBHI	CNN	Noise	No fusion
12	Nadkarni et al.	2024	Respiratory	MFCC+CS-TFT	ICBHI	CNN+XGB	Complex	No bio features
13	Jayalakshmy	2021	COPD	EMD+GTC-C	ICBHI	BiLSTM	Temporal	No spatial
14	Pham et al.	2022	Lung	Wavelet	ICBHI	CNN	Generalization	No fusion
15	Phankokkrud et al.	2025	COPD, Asthma, Pneumonia (Respiratory diseases)	MFCC feature extraction + Spectrogram + Audio Augmentation	ICBHI	CNN+LSTM	Requires large dataset and high computational cost	Need lightweight and real-time deployable models

Papers 16-20

Sl.No	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
16	Rizwana Zulfiqar et al.	2020	Lung	Spectrogram	ICBHI	CNN	Noise	No filtering
17	Sara A. Shehab et al.	2021	Respiratory	Multi-feature	ICBHI	DL	Complex	No explain
18	Haider et al.	2018	COPD	Statistical	Custom	SVM	Low acc	No DL
19	Murat Aykanat et al.	2019	Lung	MFCC	Custom	SVM	Small data	No DL
20	Fatih Demir et al.	2020	Lung	Spectrogram	ICBHI	CNN	Imbalance	No aug

Papers 21-25

Sl.No	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
21	Zeenat Tariq et al.	2020	Lung	Audio+	Custom	DL	Data need	No robust
22	Zhang et al.	2018	Respiratory	MFCC	ICBHI	LSTM	Sequence	No hybrid
23	Timothy Albiges et al.	2021	Chronic	Signal	Custom	ML	Complex	No DL
24	Panagiotis Kapetanidis et al	2022	Respiratory	Various	Multiple	Review	No impl	No unified
25	Zeenat Tariq et al.	2019	Lung	Raw	ICBHI	Dataset	Less data	Need aug

Papers 26-30

Sl.No	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
26	Rajesh Kumar Tripathy et al.	2019	Lung	Audio	Custom	ML	Low gen	No DL
27	Bahoura et al.	2006	Crackles	Wavelet	Custom	Filter	Not scale	No class
28	Zhenzhen et al.	2012	Crackles	Hilbert	Simulated	Detect	Weak	No ML
29	Reyes et al.	2008	Crackles	HHT	Clinical	Analysis	Complex	No DL
30	Graziosi Et al.	2004	Audio	CQT	Synthetic	Signal	Not lung	Gap

Paper 31,32

Sl.No	Author	Year	Disease	Method	Dataset	Model	Limitation	Gap
31	Mang et al.	2024	Lung	STFT	ICBHI	CNN	Low freq	Need CQT
			Lung	MFCC	ICBHI	DL	Detail loss	Need better
			Lung	Cochlea	ICBHI	DL	High cost	Optimize
			Lung	CQT	ICBHI	DL	Less used	Integrate
			Respiratory	Cochleogram	ICBHI	ViT	High cost	No hybrid
32	Rahul Nadkarni et al.	2024	Respiratory	Multi	ICBHI	Ensemble	Cost	Simplify

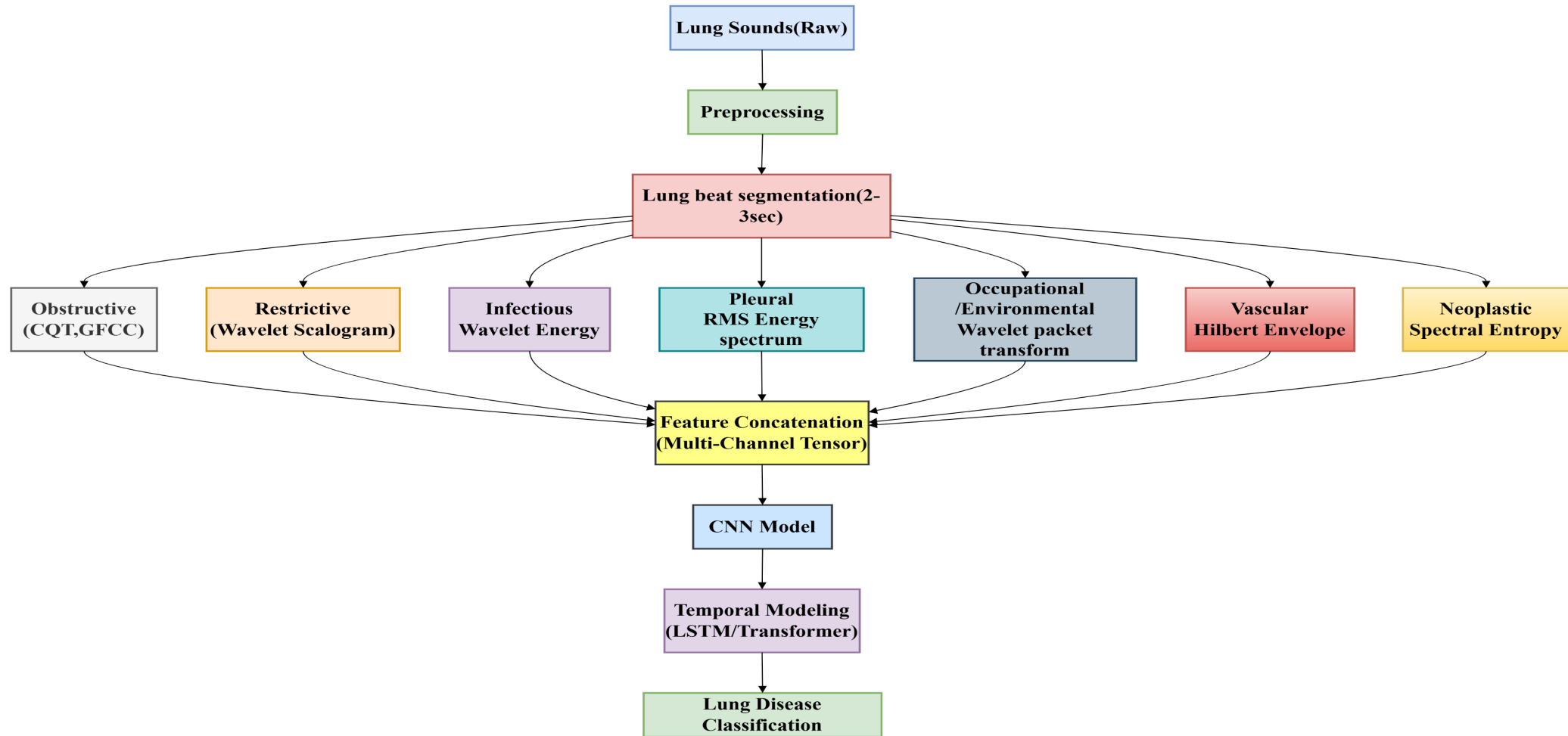
Problem statement

- Develop a lung sound analysis system for the detection, classification, and timely diagnosis of respiratory diseases.
- Focus is on identifying abnormal lung sound patterns such as wheezes and crackles associated with different diseases.
- The automated system should be able to distinguish between normal and abnormal respiratory sounds and further classify them into specific disease categories.
- It should compare extracted features with trained models to ensure accurate prediction.
- A performance output or accuracy score should be displayed to assist clinicians in decision-making.

Objectives of the work

- To design and develop an efficient lung sound classification system using advanced feature extraction techniques to enhance signal representation by using CNN.
- The performance of CNN with Support Vector Machine (SVM) and Long Short-Term Memory (LSTM) models in order to determine the most effective approach for lung sound classification
- The Comparison of CNN and LSTM for accurate classification of lung sounds to check the metrics .

Multi branch framework for lung sound analysis



Work Carried out so far

- Studied various research papers related to lung sound classification and respiratory disease detection. Analyzed different feature extraction techniques such as MFCC, STFT, CQT, and cochleogram used in existing works
- Reviewed various machine learning and deep learning models including SVM, CNN, LSTM, and Transformer-based approaches.
- Compared existing methods based on accuracy, features used, and limitations.
- Identified research gaps such as lack of feature fusion and handling of non-stationary signals.
- Formulated a proposed approach based on insights from the literature survey.

Further Work to be carried out

- Collect and augment dataset (using GAN)
- Extract features using advanced feature extraction algorithm
- Develop models (CNN, SVM, LSTM)
- Perform feature fusion
- Evaluate and optimize performance

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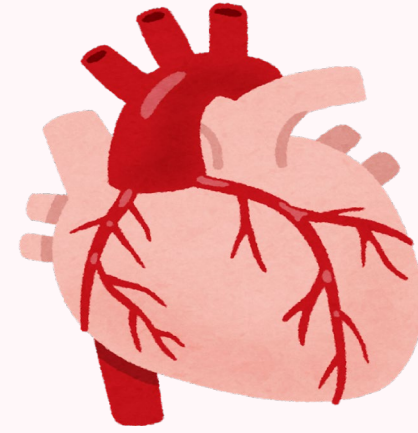
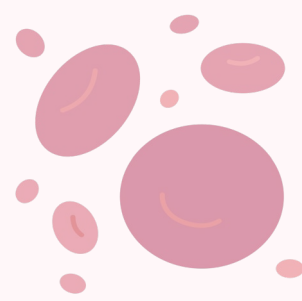
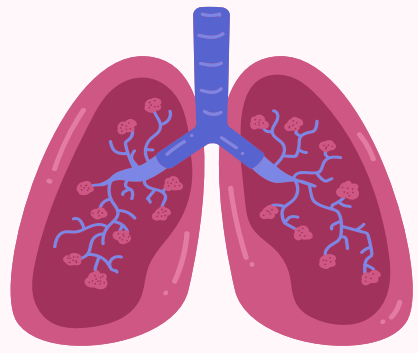
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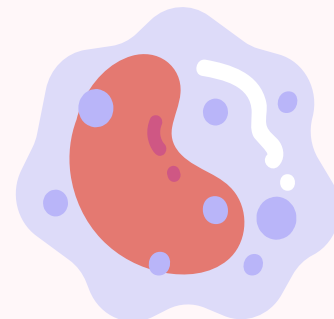
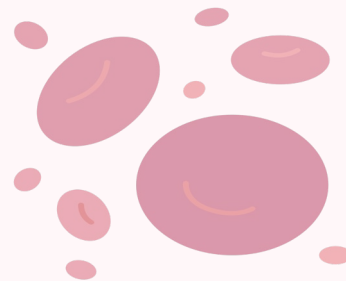
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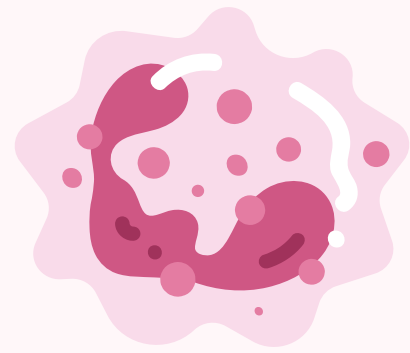
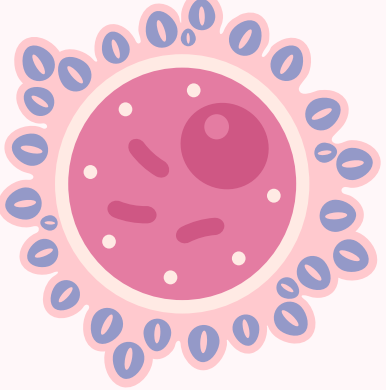
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Thank You





Welcome to Question Time

